Problem I. Sereja and Brackets

Time Limit	1000 ms
Mem Limit	262144 kB
Input File	stdin
Output File	stdout

Sereja has a bracket sequence $s_1, s_2, \ldots, s_n$, or, in other words, a string s of length n, consisting of characters "(" and ")".

Sereja needs to answer m queries, each of them is described by two integers l_i, r_i $(1 \le l_i \le r_i \le n)$. The answer to the i-th query is the length of the maximum correct bracket subsequence of sequence $s_{l_i}, s_{l_i+1}, \ldots, s_{r_i}$. Help Sereja answer all queries.

You can find the definitions for a subsequence and a correct bracket sequence in the notes.

Input

The first line contains a sequence of characters $s_1, s_2, \ldots, s_n$ $(1 \le n \le 10^6)$ without any spaces. Each character is either a "(" or a ")". The second line contains integer m $(1 \le m \le 10^5)$ — the number of queries. Each of the next m lines contains a pair of integers. The i-th line contains integers l_i, r_i $(1 \le l_i \le r_i \le n)$ — the description of the i-th query.

Output

Print the answer to each question on a single line. Print the answers in the order they go in the input.

Examples

Input	Output
())(())(())(7 1 1 2 3 1 2 1 12 8 12 5 11 2 10	0 0 2 10 4 6 6

Note

A *subsequence* of length $|x|$ of string $s = s_1 s_2 \dots s_{|s|}$ (where $|s|$ is the length of string s) is string $x = s_{k_1} s_{k_2} \dots s_{k_{|x|}}$ $(1 \le k_1 < k_2 < \dots < k_{|x|} \le |s|)$.

A *correct bracket sequence* is a bracket sequence that can be transformed into a correct aryphmetic expression by inserting characters "1" and "+" between the characters of the string. For example, bracket sequences "()()", "(())" are correct (the resulting expressions "(1)+(1)", "((1+1)+1)"), and ")(" and "(" are not.

For the third query required sequence will be «()».

For the fourth query required sequence will be «()(())(())».